CS5187: Vision & Image

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Tutorial 04: Classification and depth estimation

（1）Classification

What we will cover today:

Introduction to the CIFAR-10 dataset.

Building a custom Convolutional Neural Network (CNN) from scratch.

Adapting a state-of-the-art Vision Transformer (ViT) for a custom task.

Understanding the differences in data processing for CNNs vs. ViTs.

Writing a standard PyTorch training loop.

（1）Classification

What are we trying to classify?

The Dataset: 60,000 color images (50,000 for training, 10,000 for testing).

The Classes: 10 distinct categories (airplanes, cars, birds, cats, deer, dogs, frogs, horses, ships, trucks).

The Challenge: The images are incredibly small—only **32x32 pixels**.

Why it matters: The low resolution forces our neural networks to learn highly efficient feature representations to tell a dog apart from a cat.

（1）Classification

CNN

•**How it works:** Uses sliding windows (filters/kernels) to detect local patterns like edges, corners, and textures.

•**Architecture:** Layers of Convolutions $\rightarrow$ Pooling (downsizing) $\rightarrow$ Fully Connected (Classification) Layers.

•**Why it fits CIFAR-10:** CNNs are naturally suited for 32x32 images. They are computationally lightweight and train quickly from scratch on smaller datasets.

（1）Classification

The Modern State-of-the-Art

How it works: Inspired by Natural Language Processing. It treats an image like a sentence.

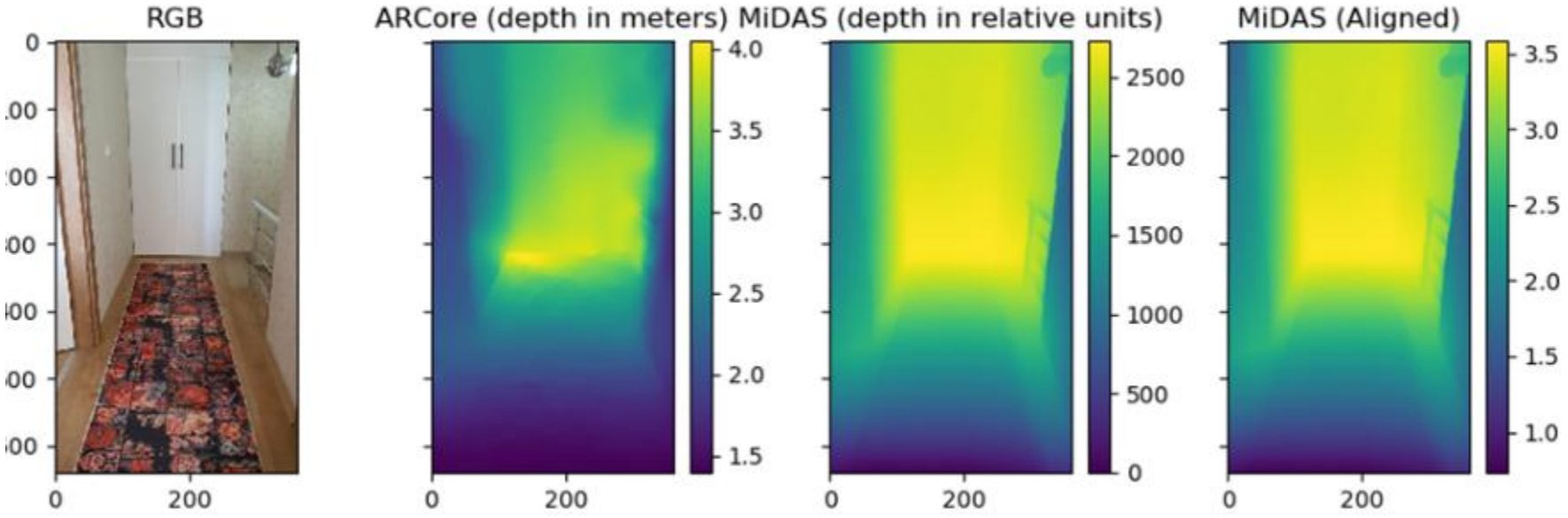
Patching: The image is chopped into a grid of smaller "patches" (usually 16x16 pixels).

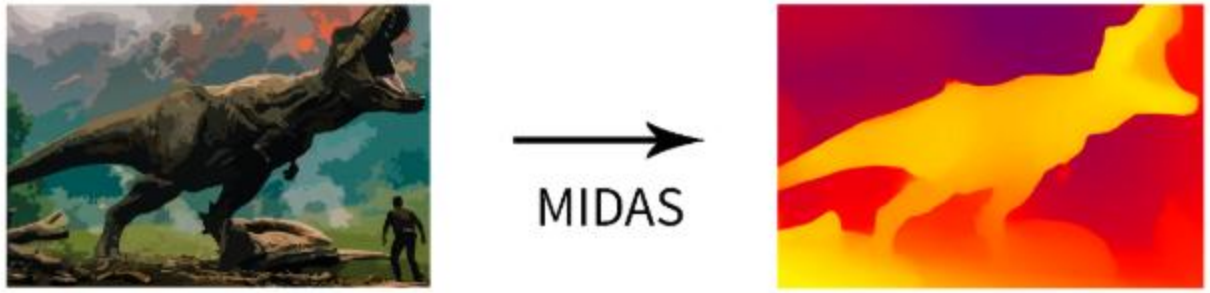
Attention: Instead of local filters, ViTs use "Self-Attention" to look at the entire image at once and see how every patch relates to every other patch.

（1）Classification

Refer to the supplemented code

（2）Neural Network Depth Estimation





（2）Neural Network Depth Estimation

Refer to the supplemented code